

ENGR098 LAB 10

November 6, 2025

1 Objective

The objective of this lab is to design, simulate, and analyze a Wien-Bridge Oscillator using a UA741 operational amplifier in PSpice. The goal is to observe sustained oscillations and measure the output frequency by adjusting resistor and capacitor values.

2 Circuit Design

The Wien-Bridge Oscillator is an RC feedback oscillator that generates a sinusoidal waveform without an external input signal. The circuit consists of:

- An operational amplifier (UA741)
- Resistors: R_1 , R_2 , R_f , R_g
- Capacitors: C_1 , C_2
- Dual DC power supply (± 15 V)

3 Theory and Formula

The frequency of oscillation for the Wien-Bridge Oscillator is given by:

$$f = \frac{1}{2\pi RC}$$

where:

- $R = R_1 = R_2$
- $C = C_1 = C_2$

For stable oscillation, the amplifier's gain must satisfy:

$$\frac{R_f}{R_g} = 2$$

This ensures that the feedback provides the correct amplitude without distortion.

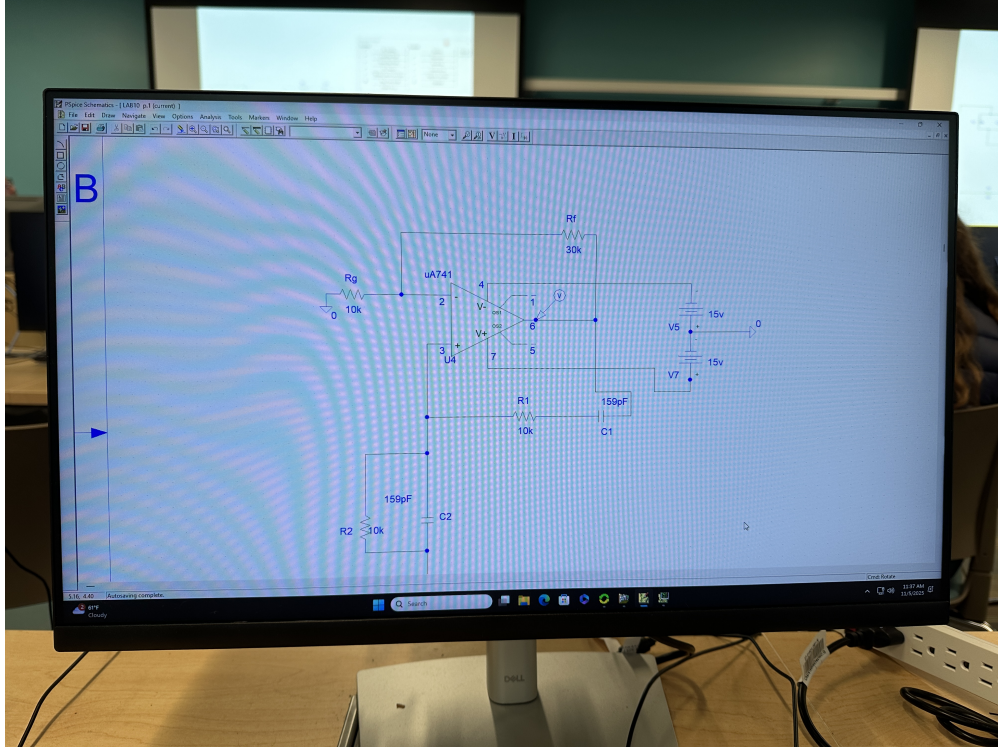


Figure 1: Initial Wien-Bridge Oscillator Circuit ($C_1=C_2=159$ pF, $R_1=R_2=10$ k Ω)

4 Simulation and Results

4.1 Case 1: Small Capacitance ($C_1 = C_2 = 159$ pF)

The circuit was simulated with $R_1 = R_2 = 10$ k Ω and $C_1 = C_2 = 159$ pF. The expected frequency was:

$$f = \frac{1}{2\pi(10 \times 10^3)(159 \times 10^{-12})} \approx 100 \text{ kHz}$$

4.2 Case 2: Larger Capacitance ($C_1 = C_2 = 0.1$ μ F)

When both capacitors were increased to 0.1 μ F and resistors adjusted ($R_1 = R_2 = 80$ k Ω , $R_f = 20$ k Ω , $R_g = 10$ k Ω), the expected frequency decreased:

$$f = \frac{1}{2\pi(80 \times 10^3)(0.1 \times 10^{-6})} \approx 20 \text{ Hz}$$

5 Analysis

The simulation confirms that:

- The frequency of oscillation decreases as either R or C increases.

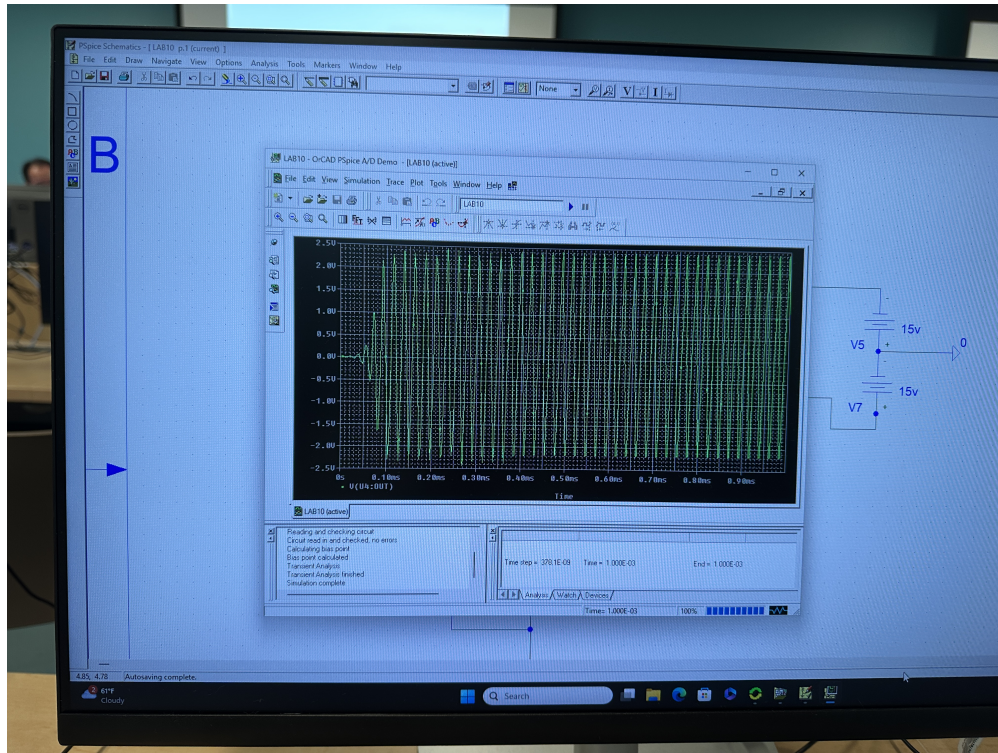


Figure 2: Simulation output showing high-frequency oscillation near 100 kHz.

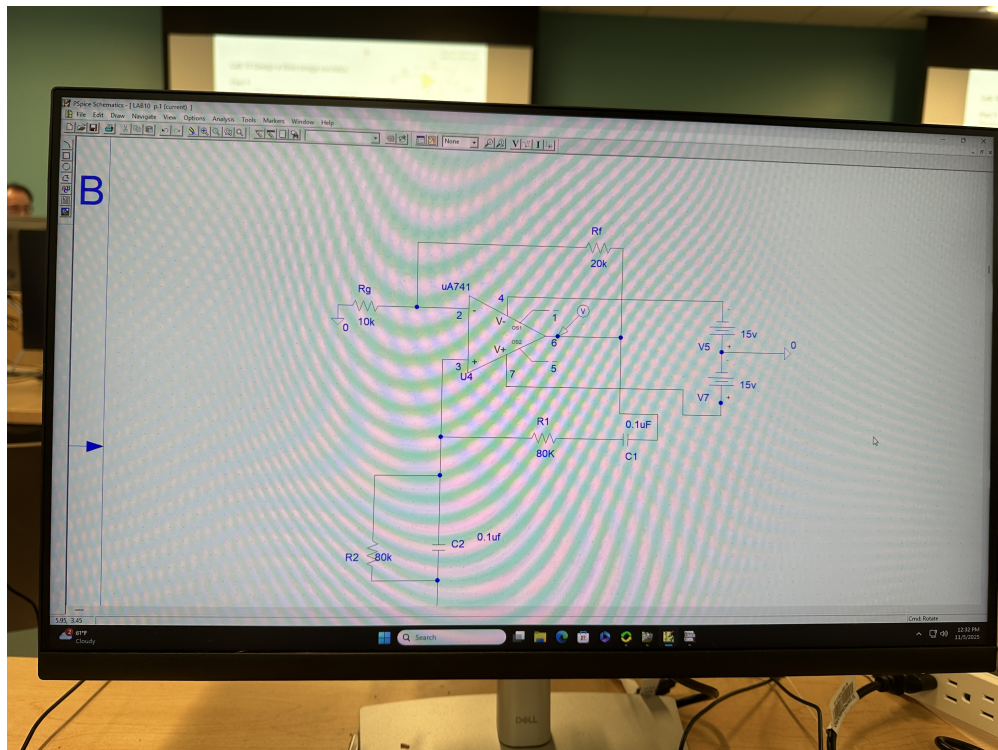


Figure 3: Simulation output showing low-frequency sinusoidal oscillation around 20 Hz.

- The UA741 op-amp can generate stable oscillations when the gain condition $\frac{R_f}{R_g} \approx 2$ is met.
- The circuit produces a clean sine wave with larger capacitors and higher resistance.

The output amplitude remains limited by the op-amp's supply voltage (± 15 V), and slight distortion may occur if the gain is too high.